Project Checkpoint 2 📊

Due Mon, Nov 16 @ 11:59pm

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In Checkpoint 2, you turn your Checkpoint 1 outcome into a yes/no outcome, fit logistic regressions, and use model selection or regularization with cross-validation to choose the best predictors. You can make a copy of this document (File > Make a copy) for your own project planning. This document looks super long, but that's because we provide a step-by-step checklist for what your project should include!

Overview

- Submit (one per group): a link to a recorded Zoom video of 7-10 minutes
    - Slide template: Project Checkpoint 2 — Slide Template. You may customize your slides however you like, as long as they have all the necessary information.
- Submit: Your group project video and Google Drive folder of code notebooks
- Contribution: Your group keeps its Checkpoint 1 dataset and train/test split. Together, you turn your outcome into a binary (0/1) outcome. Each member then uses a different method to choose the best predictors from all the predictors in your dataset with k-fold cross-validation, reports AIC and BIC, interprets their selected logistic regression, and evaluates it on the test set. Then your group compares the models.
    - We are providing a code notebook template: COGS109_CP2_Notebook_Template.ipynb and example output: COGS109_CP2_Notebook_Template_example_output.ipynb. You DO NOT have to use these notebook templates—they are just provided to help you.
- Points: 3 total, with 0.75 for the group's shared slides and 2.25 for your own model. Rubric at the end of this page.
- Late policy: 24-hour grace period, then -0.2 points per day, with a hard cutoff at 11:59pm on Friday, November 20.

Checkpoint 2 Components:

1. As a group, turn your Checkpoint 1 outcome into a binary (0/1) outcome and check how balanced the two classes are.
2. Keep your Checkpoint 1 train/test split. Split the training set into k folds that serve as validation sets. Your candidate predictors are all the predictors in your dataset, leaving out any text or date/time columns.
3. Each member uses a different method to choose the best predictors from that pool with k-fold cross-validation, and reports AIC, BIC, and CV error rate.
4. Each member fits their selected logistic regression on the training set and interprets its coefficients as odds ratios.
5. Each member evaluates their final model on the test set and reports test accuracy, the confusion matrix, precision, recall, and the ROC curve with its AUC.

6. The group compares the models and records the video.

Step 0: Code notebooks

- [ ] We are providing a **code notebook template:** COGS109_CP2_Notebook_Template.ipynb and **example output:** COGS109_CP2_Notebook_Template_example_output.ipynb
    - [ ] You DO NOT have to use these notebook templates—they are just provided to help you get started. They include every step that is in this document.
    - [ ] You may also use snippets from your Checkpoint 1 notebook and the code notebooks provided in the homeworks.

Step 1: Turn your outcome into a binary outcome

Logistic regression predicts a yes/no outcome, so your group turns its Checkpoint 1 outcome into two classes.

- [ ] **Keep your Checkpoint 1 dataset and outcome.** Choose a threshold that splits the outcome into two classes: 1 = above the threshold, 0 = at or below it.
    - For example, if you predicted exam scores in Checkpoint 1, your new outcome is whether a student scored above 50% (1) or not (0).
    - As another example, if you predicted scores on a symptom scale where a score above 80 means a diagnosis, your new outcome is whether a person is diagnosed (1 = scored above 80) or not (0 = scored 80 or below). Here the classes won't be split 50/50, because most people may score 80 or below.
- [ ] **Choose a threshold that means something to your audience,** such as a passing grade, a health cutoff, or a price point. If there is no natural cutoff, use the median of the training set so that it will be split roughly 50/50.
- [ ] **Check the class balance:** what proportion of the training set is 1? Aim for at least 20% of rows in each class. If one class is very rare, be prepared to address class imbalance using the methods we learned in class.
- [ ] **Do NOT use the original continuous outcome as a predictor in your model.** It would give away the answer.
- [ ] **Rewrite your modeling question for the binary outcome.** For example, *"What factors predict whether a student scores above 50%?"*

Step 2: Keep your test set and set up your validation folds

- [ ] **Use the same train/test split as Checkpoint 1** (the same proportions and random seed). The test set stays untouched until Step 5.
- [ ] **Split the training set into k folds** (k = 5 or 10), using one random seed for the whole group. In k-fold cross-validation, each fold takes a turn as the **validation set** while the model is fit on the other k − 1 folds. The average error across the k validation sets is your **CV error rate**.
    - [ ] Everyone uses the same folds. Otherwise your CV errors can't be compared.
    - [ ] Use the **CV error rate** (the proportion of validation cases your model misclassifies, using a 0.5 probability cutoff) to compare models.

- [ ] **Agree on your pool of candidate predictors:** all the predictors in your dataset, leaving out any text or date/time columns and the original continuous outcome. Every member starts from this same pool.

Step 3: Choose your predictors with cross-validation

In Checkpoint 1, you chose your predictors by hand. In Checkpoint 2, you let a **model selection method choose them.** Each member starts from the group's **pool of all predictors in the dataset (leaving out any text or date/time columns)** and chooses **one method** from the table below. **Each group member must choose a different method,** so no two members in your group use the same one.

Keep in mind that in the lectures, we used **linear regression** as an example for introducing these model selection and regularization methods. However, all of these methods can also be used in logistic regression to select predictors!

Method	How it works	What cross-validation chooses
Forward stepwise selection	Starts with no predictors and adds the one that improves the fit most at each step	The number of predictors
Backward stepwise selection	Starts with all candidates and removes the least useful one at each step	The number of predictors
Lasso (L1) logistic regression	Penalizes the size of the coefficients, which shrinks some of them to exactly 0	The penalty strength λ, and so which predictors stay in
Ridge (L2) logistic regression	Penalizes the size of the coefficients, which shrinks them all toward 0 but keeps every predictor in the model	The penalty strength λ, and so how much to shrink
Elastic net logistic regression	Mixes the lasso and ridge penalties, which helps when predictors are correlated	The penalty strength λ and the mix between lasso and ridge

Check how many predictors (p) you have compared with training rows (n) before you choose a method; count each dummy variable as a predictor. **If p is larger than n,** backward stepwise selection won't work, because it starts from the full model, which can't be fit. Use forward stepwise selection, lasso, ridge, or elastic net. **If p is close to n,** the full model is unstable, so backward stepwise is unreliable; lasso, ridge, and elastic net are the safest choices.

- [ ] **Use only the training set,** with the group's shared k folds. Never use the test set to choose a model.
- [ ] **For lasso, ridge, or elastic net, standardize your quantitative predictors first,** so the penalty treats them equally.
- [ ] **Plot your CV error rate** against the number of predictors (stepwise) or against $\log(\lambda)$ (lasso, ridge, elastic net). Mark the value you chose.
- [ ] **Stepwise: also compute AIC and BIC** for the best model of each size, and plot them on the same x-axis as your CV error rate. Which size does each measure pick?
- [ ] **List the predictors your method kept.** Ridge keeps every predictor, so list the ones with the largest standardized coefficients.

Both AIC and BIC reward a good fit and penalize extra predictors; BIC penalizes them more, so it often picks a smaller model. Lower is better for both.

$$\text{AIC} = -2 \log L + 2d \qquad \text{BIC} = -2 \log L + d \log n$$

Here L is the model's likelihood on the training set, d is the number of coefficients, and n is the number of training rows.

Step 4: Compare and interpret your selected model

Your **selected model** is the model your method chose in Step 3. For stepwise, lasso, or elastic net, refit an **ordinary logistic regression** on the training set using just the predictors your method kept. For ridge, use the ridge fit itself.

- ☐ **Report your selected model's AIC, BIC** (defined in Step 3), **and CV error rate,** so your group can compare every member's regularized model on the same measures in Step 6. Lower is better for all three.
- ☐ **Ridge:** AIC and BIC don't apply to a penalized fit, so report only the CV error rate for ridge.
- ☐ **Write your final model on your slides in math,** in log-odds form, using your actual variable names. If your model has too many predictors to fit, write the log-odds outcome and the **two predictors with the largest coefficients** (largest in absolute value, using standardized coefficients so they're comparable), then end with "+ …". For example, if lasso kept price, advertising budget, and shelf location (baseline: Bad) in a model of whether a car seat store has high sales (1 if sales are above 8,000 units), your model would be:

$$\log(p / (1 - p)) = \beta_0 + \beta_1 \cdot \text{Price} + \beta_2 \cdot \text{Advertising} + \beta_3 \cdot \text{ShelveLoc_Good} + \beta_4 \cdot \text{ShelveLoc_Medium}$$

- ☐ Say what p is, for example, p = P(High sales = 1). Each dummy variable gets its own term, and the baseline category has none.
- ☐ **Interpret your coefficients as odds ratios** (e^β), holding the other predictors fixed:
  - ☐ **Interpret at least two coefficients, either quantitative OR qualitative:** for a quantitative predictor, how the odds of the outcome change for a one-unit increase in that predictor. For example, if $e^{\beta_1} = 0.90$, each $1 increase in price multiplies the odds of high sales by 0.90 (10% lower odds). For a qualitative predictor, how the odds for that category compare with the baseline category; name the baseline.
  - ☐ **Ridge:** your coefficients are shrunk toward 0, so say that your odds ratios are pulled toward 1.
- ☐ Optional: use the bootstrap to get a 95% confidence interval for one of your odds ratios.

Step 5: Test your final model

- ☐ Use your final model to predict the **test set**, once. Classify a case as 1 if its predicted probability is above 0.5.
- ☐ Report:
  - ☐ **Test accuracy:** the proportion of test cases classified correctly. Higher is better.
  - ☐ **Confusion matrix:** the counts of true positives, false positives, true negatives, and false negatives.
  - ☐ **Precision** (the proportion of predicted 1s that are truly 1) and **recall** (the proportion of true 1s predicted as 1).
  - ☐ **ROC curve and AUC:** plot the ROC curve on the test set, which shows the true positive rate (recall) against the false positive rate (the proportion of true 0s predicted as 1) as the probability cutoff moves from 0 to 1. Report the **AUC** (area under the curve) and interpret it in one sentence. The AUC is the probability that your model gives a randomly chosen 1 a higher predicted probability than a randomly chosen 0: 0.5 is no better than chance, and 1 is perfect.

For example, "An AUC of 0.82 means that, for a random diagnosed and undiagnosed person, our model gives the diagnosed person the higher probability 82% of the time."

- [ ] **Compare with the baseline:** the accuracy of always predicting the most common class in the training set. Your model should beat it.
- [ ] Remember: the point of the course project is **not necessarily to maximize the accuracy of your predictions**, but rather to learn about how to follow the **entire predictive modeling workflow correctly.** Don't worry if your prediction accuracy or your other model evaluation metrics are not very good!

Step 6: Compare your models as a group

- [ ] Put every member's results in one table on the last slide: method, number of predictors, CV error rate, AIC, BIC, test accuracy, AUC, precision, and recall, plus each member's full confusion matrix (true positives, false positives, true negatives, and false negatives). Precision is the proportion of predicted 1s that are truly 1; recall is the proportion of true 1s predicted as 1. Add a column for the baseline.
- [ ] Decide which model predicts best on the test set, and explain why in one or two sentences. Did the methods agree on which predictors matter? Did AIC, BIC, and CV point to the same model?

Slides and video

- [ ] **Make a copy** of the Project Checkpoint 2 — Slide Template (File → Make a copy) and replace it with your group's content.
- [ ] **Length:** 7-10 minutes.
- [ ] Each member **presents their own model slides** in their own voice. Anyone can present the group slides.

Slide 1: Binary outcome and question (anyone, 45 sec)

- [ ] Group number and member names
- [ ] Dataset name and source
- [ ] Your Checkpoint 1 outcome, your threshold, and what 1 and 0 mean
- [ ] The class balance: the proportion of 1s in the training set
- [ ] Your modeling question for the binary outcome, in one sentence

Slide 2: Workflow (anyone, 45 sec)

- [ ] Your train/test split (the same as Checkpoint 1): proportions and random seed
- [ ] Your validation folds: k and the random seed
- [ ] Your shared candidate pool: how many predictors, split into quantitative and qualitative
- [ ] Which member used which selection method

Member model selection slide (each member, 1 min)

- [ ] Your name and your selection method
- [ ] The plot of CV error rate against the number of predictors (stepwise) or log(λ) (lasso, ridge, elastic net), with your chosen value marked. Stepwise: AIC and BIC on the same plot
- [ ] The predictors your method kept (ridge: the predictors with the largest standardized coefficients)
- [ ] Your selected model's AIC, BIC, and CV error rate (ridge: CV error rate only)

Member selected model slide (each member, 1 min 15 sec)

- [ ] Your selected model written in math in log-odds form, with p defined, and the fitted equation with your coefficient estimates. If there are many predictors, show the two with the largest coefficients, then "+ …"
- [ ] Your interpretation of at least two coefficients as odds ratios
- [ ] Your test accuracy, confusion matrix, precision, and recall, next to the baseline accuracy
- [ ] Your ROC curve on the test set, with the AUC interpreted in one sentence

Repeat the member slides for each member.

Last slide: Model comparison (anyone, 45 sec)

- [ ] A table with one column per member, plus the baseline: method, number of predictors, CV error rate, AIC, BIC, test accuracy, AUC, precision, recall, and the confusion matrix (TP/FP/TN/FN)
- [ ] Which model predicts best, and why, in one or two sentences
- [ ] Whether AIC, BIC, and CV pointed to the same models

Presentation and slide tips

- [ ] Write each title as the slide's takeaway, for example, "Lasso kept 4 of 9 predictors."
- [ ] One main idea per slide, don't write paragraphs of text, and use at least 14pt font.
- [ ] Label every axis in your figures!
- [ ] When you present a figure, say what's plotted first on the y and x axis, then what to take away from it.

Recording and submitting

- [ ] **Slides:** we recommend Google Slides for making slides together. Make a copy of the Project Checkpoint 2 — Slide Template and share it with your group.
- [ ] **Recording:** we recommend that you all join one Zoom room and use the record meeting function to get a file of your video. You may also record separately and splice the clips together.
- [ ] **Uploading:** upload the recording as an "Unlisted" video on one member's personal YouTube account. Open the link in a private browser window to check that it plays.
- [ ] **Notebooks:** make a Google Drive folder with one code notebook per group member, and put that member's name in each file name. Share the folder so anyone with the link can view it.
- [ ] **Submitting (one per group):** one member submits the YouTube link and the Drive folder link through the Checkpoint 2 form.
- [ ] Help: TAs are always available to help during their office hours or Friday discussion section. Reach out often and early!

Grading rubric

Project Checkpoint 2 is out of 3 points. Use this rubric as your checklist before you submit.

Part	Item	Points	Full credit means
Group	Binary outcome	0.25	• The threshold is explained, with what 1 and 0

	and workflow		mean. ● The class balance is reported. ● The test set is the same as in Checkpoint 1, and the k folds and candidate pool are described. ● The group stays within the time limit.
Group	Model comparison	0.5	● One table with every member's CV error rate, AIC, BIC, test accuracy, AUC, precision, recall, and confusion matrix, on the same test set, plus the majority-class baseline. ● The best model is identified with a clear reason.
Individual	Logistic model in math	0.25	● The selected model is written correctly in log-odds form, with p defined, and the fitted equation is shown.
Individual	Interpretation	0.5	● At least two coefficients interpreted correctly as odds ratios, with the baseline named for any qualitative predictor. ● The test ROC curve is shown, and the AUC is interpreted correctly.
Individual	Model selection	0.75	● Uses a different method from teammates. ● Cross-validation uses only the training set and the shared folds. ● The CV plot marks the chosen value, and the kept predictors are listed. ● The selected model's AIC, BIC, and CV error rate are reported (ridge: CV error rate only).
Individual	Slide quality and clarity	0.5	● Slides are clear and not cluttered. ● Takeaway titles with one main idea per slide. ● 14pt or larger. ● All figures have labeled axes with units, and each figure is explained.
Individual	Code notebook	0.25	● Code notebook is included and shows all plots and analyses used in the presentation.